



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO P.B.SAKTHIVEL & COMPANY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

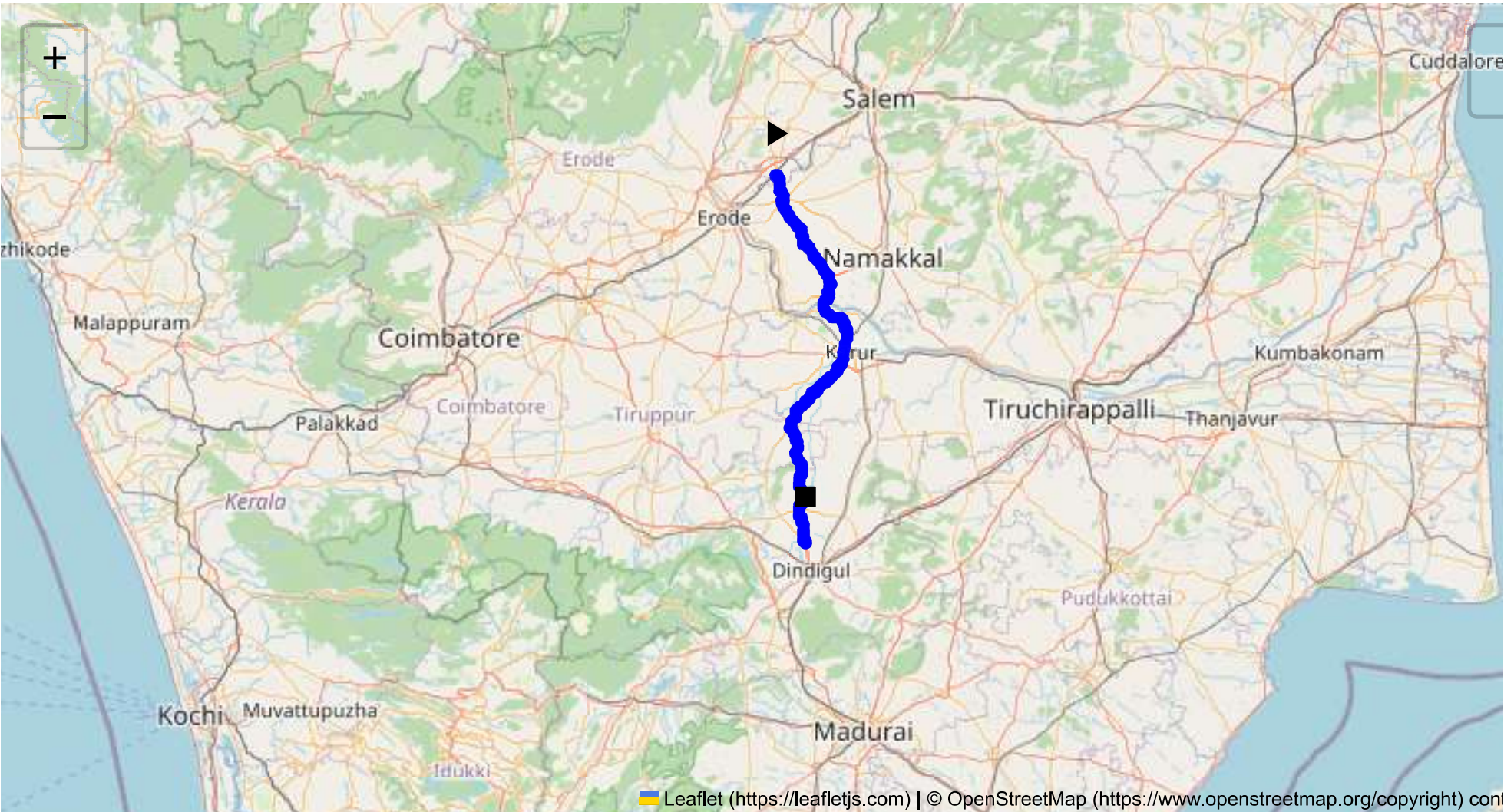
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 133.05 km
Estimated Duration: 2.3 hours
Adjusted Duration (Heavy Vehicle): 2.8 hours
Start: (11.4381, 77.8734)
End: (10.444745, 77.949501)



Welcome to the Journey Risk Management Study

To analyze the route from CVQF+23W, Sangam, Tamil Nadu 637301, India to CWVX+JH5, Tamil Nadu 624709, here is a comprehensive report:

1. Overview of the Route Map

The route primarily utilizes National Highways and major state roads, passing through towns and rural areas. The journey is expected to involve various intersections and bypasses around populated areas.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Conditions:** This region generally experiences a tropical climate with moderate temperatures, high humidity, and distinct monsoon seasons.
- **Hazards:** During the monsoon (June to September), heavy rains can lead to flooding and reduced visibility. Tropical storms are also possible, affecting driving conditions.

3. Traffic Patterns

- **Peak Hours:** Congestion is likely during morning (8:00-10:00 AM) and evening (5:00-7:00 PM) rush hours, especially near urban centers.
- **Congestion Areas:** Intersections near major towns such as Salem and Namakkal could see significant delays due to local traffic.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** The National Highways are generally in good condition with regular maintenance. However, some state roads, especially in rural stretches, may have potholes and uneven surfaces.
- **Infrastructure:** Adequate signage and rest stops are expected along the highways, but less so on smaller connecting roads.

5. Alternative Routes for Emergencies

- If the primary route is blocked, detour options include smaller state roads connecting between National Highways NH 44 and NH 83. Close attention should be paid to road quality and traffic conditions on these alternate routes.

6. Local Regulations Affecting Hazardous Material Transport

- Regulations require appropriate documentation for transporting hazardous materials. Specific routes may be designated for such vehicles, and night-time travel could involve additional restrictions or curfews near populated areas to minimize risk.

7. Historical Incidents

- Review of available data shows occasional historical incidents involving heavy vehicles, primarily due to driver error or mechanical failures. Instances of hazardous material spills are rare but have occurred, highlighting the importance of strict adherence to safety protocols.

8. Environmental Considerations and Sensitive Areas

- The route may pass near protected areas or agricultural land. It is crucial to avoid contamination of water bodies and soil, especially large rivers and lakes in the region.

9. Communication Coverage

- Overall, mobile network coverage is decent along major highways, with occasional dead zones in rural stretches. Reliable communication devices and network providers should be considered essential.

10. Estimated Emergency Response Times

- Emergency services in urban and semi-urban areas can typically reach incidents within 15 to 30 minutes. In rural stretches, response times could extend to an hour or more, depending on accessibility.

11. Overall Summary of Risk Assessment

This route encompasses moderate risk for transporting hazardous materials mainly due to weather conditions and potential traffic congestion. Proper planning, strict adherence to safety regulations, and awareness of local regulations and environmental considerations can mitigate most transportation hazards. Preparedness for communication blackouts and ensuring vehicles are well-maintained can further enhance trip safety.

Given the complexities associated with transporting hazardous materials, it is imperative to conduct pre-trip safety checks, provide driver training on emergency protocols, and maintain unrestricted communication with dispatch and emergency services.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44019, 77.87547	15 KM/Hr
3	Turn	Medium	11.39685, 77.88861	30 KM/Hr
4	Turn	Medium	11.37902, 77.89500	30 KM/Hr
5	Turn	Medium	11.37850, 77.89453	30 KM/Hr
6	Turn	Medium	11.16950, 78.01544	30 KM/Hr
7	Turn	Medium	11.16764, 78.01565	30 KM/Hr
8	Turn	Medium	11.15546, 78.02170	30 KM/Hr
9	Turn	High	11.14521, 78.02239	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	High	11.14480, 78.02291	15 KM/Hr
11	Turn	Medium	11.10102, 78.00572	30 KM/Hr
12	Turn	Medium	11.08294, 78.00335	30 KM/Hr
13	Turn	Medium	11.05261, 78.05138	30 KM/Hr
14	Turn	Medium	10.92818, 78.05335	30 KM/Hr
15	Turn	Medium	10.89913, 78.03032	30 KM/Hr
16	Turn	Medium	10.79857, 77.92714	30 KM/Hr
17	Turn	Medium	10.78576, 77.92670	30 KM/Hr
18	Turn	Medium	10.75496, 77.91210	30 KM/Hr
19	Turn	Medium	10.64356, 77.94370	30 KM/Hr
20	Turn	Medium	10.50627, 77.94250	30 KM/Hr
21	Turn	Medium	10.44123, 77.95234	30 KM/Hr
22	Turn	Medium	10.43831, 77.95397	30 KM/Hr
23	Blind Spot	Blind Spot	10.43776, 77.95400	10 KM/Hr
24	Turn	Medium	10.44111, 77.95206	30 KM/Hr
25	Turn	Medium	10.44248, 77.94994	30 KM/Hr
26	Turn	High	10.44255, 77.94900	15 KM/Hr
27	Turn	High	10.44474, 77.94888	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
9	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
10	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
19	hospital	Government Hospital, Thadicombu	10.4384149, 77.9550422	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
18	college	valluvar catering	10.9553473, 78.0606583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44019, 77.87547



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39685, 77.88861



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37902, 77.89500



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16950, 78.01544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16764, 78.01565



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.15546, 78.02170



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14521, 78.02239



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.10102, 78.00572



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.08294, 78.00335



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.05261, 78.05138



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.92818, 78.05335



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.89913, 78.03032



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.79857, 77.92714



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.78576, 77.92670



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.75496, 77.91210



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.64356, 77.94370



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.50627, 77.94250



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.44123, 77.95234



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.43831, 77.95397



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 10.43776, 77.95400



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.44111, 77.95206



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.44248, 77.94994



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.44255, 77.94900



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.44474, 77.94888

